

VBD (Variable Buoyancy Device)

The VBD is the Seaglider's buoyancy engine and the reason it moves at all: a hydraulic system in the aft endcap that moves low-viscosity oil between an **internal reservoir** inside the pressure hull and an **external bladder** outside the hull (but inside the fairing). Pumping oil out increases the glider's displaced volume without changing its mass — it gets lighter than the water and climbs; bleeding oil back in shrinks it and it sinks. It is also the single largest energy consumer on the vehicle, so most piloting-for-endurance decisions are ultimately VBD decisions.

Source

Paraphrased from the APL-UW IOP SGX *Documentation* (v1.0, 2024), the manufacturer's air-bleed procedure (support correspondence, 2018), the community "Cycletron in Pupa" pump-cycling notes, IOP webinar/office hours material, and 2018–2024 field and refurbishment correspondence between Seaglider operators, APL-UW IOP and service providers (VBD rebuilds, seal failures), and 2017–2023 manufacturer support correspondence (cycletron analysis, cold-water pump rates). Hardware details vary between SG, SGX, and Deepglider variants and build years — defer to APL-UW IOP and your glider's documentation.